

IRVINE AGCM, AB, Canada

WMO: 717950

Lat: 49.9892N

Lon: 110.2614W

Elev: 795

StdP: 92.13

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-29.0	-25.9	-32.4	0.2	-28.8	-29.1	0.3	-25.8	13.8	-0.4	12.4	-1.8	2.7	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	32.8	16.3	30.8	16.2	28.6	15.8	18.6	26.9	17.7	26.5	16.8	26.0	5.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	12.4	21.7	14.7	11.5	20.2	13.6	10.7	19.4	56.0	26.7	52.9	26.5	50.1	26.0	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	11.3	10.0	DB	-33.9	36.2	2.3	1.7	-35.6	37.4	-36.9	38.3	-38.2	39.3	-39.8	40.5	(4)
(5)			WB	-34.1	20.7	2.2	1.2	-35.7	21.6	-37.0	22.3	-38.2	23.0	-39.8	23.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-6.9	-8.3	-0.5	5.8	11.6	16.0	19.6	19.1	14.1	6.4	-0.7	-7.9	(6)	
	DBStd	11.91	8.78	8.74	8.14	5.14	4.61	3.61	3.22	3.86	5.17	5.62	7.68	8.91	(7)	
	HDD10.0	2585	525	513	330	143	36	2	0	0	18	139	324	555	(8)	
	HDD18.3	4753	783	746	585	376	214	88	22	37	147	371	570	813	(9)	
	CDD10.0	1038	0	0	3	17	85	182	298	283	139	27	4	0	(10)	
	CDD18.3	166	0	0	0	0	5	17	63	61	19	1	0	0	(11)	
	CDH23.3	2456	0	0	0	8	101	264	902	870	297	14	0	0	(12)	
	CDH26.7	905	0	0	0	2	17	79	345	347	113	2	0	0	(13)	
(14) Wind	WSAvg	4.6	4.8	4.3	4.2	5.1	4.3	4.7	4.2	3.9	4.1	5.3	5.0	5.1	(14)	
(15) Precipitation	PrecAvg	390	20	16	20	36	55	79	37	34	32	24	19	18	(15)	
	PrecMax	591	39	39	37	66	95	173	146	88	63	59	44	39	(16)	
	PrecMin	230	6	3	3	11	10	19	5	8	1	5	4	3	(17)	
	PrecStd	72	8	9	10	15	22	37	29	21	17	13	9	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.7	18.4	23.7	28.9	32.7	35.1	35.0	33.2	25.3	19.6	11.9	(19)	
		MCWB	4.7	6.0	8.7	11.5	14.3	16.4	17.2	16.8	15.4	13.6	9.5	5.6	(20)	
	2%	DB	8.3	7.6	15.4	20.9	26.2	28.7	32.7	32.6	30.1	21.0	14.8	8.8	(21)	
		MCWB	3.9	3.2	7.2	10.1	13.6	16.0	16.9	16.2	15.1	11.2	7.7	3.8	(22)	
	5%	DB	6.4	5.2	12.5	17.8	23.8	26.2	30.4	30.7	26.8	18.2	12.1	6.3	(23)	
		MCWB	3.1	1.9	5.9	8.4	12.2	15.5	16.9	15.9	14.0	10.0	6.2	2.6	(24)	
	10%	DB	4.7	3.3	9.6	15.1	21.1	24.0	28.3	28.3	23.6	15.4	9.5	3.9	(25)	
		MCWB	1.7	0.7	4.3	6.9	11.1	14.2	16.3	15.5	12.9	8.4	4.6	1.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	6.3	9.2	12.3	15.2	19.2	20.8	19.0	17.2	14.0	9.9	6.4	(27)	
		MCDB	10.3	11.5	16.9	22.9	26.0	25.4	29.9	27.9	27.1	23.6	18.0	11.1	(28)	
	2%	WB	4.1	3.4	7.7	10.4	14.1	17.4	19.2	18.0	15.8	12.2	8.1	4.3	(29)	
		MCDB	7.7	6.9	14.5	19.8	24.2	26.3	27.4	27.0	27.5	19.2	14.1	8.3	(30)	
	5%	WB	3.0	2.1	6.1	8.9	13.1	16.2	18.1	17.1	14.6	10.4	6.4	2.6	(31)	
		MCDB	6.4	5.2	11.9	16.8	22.0	24.3	26.3	27.4	24.9	16.7	11.6	6.1	(32)	
	10%	WB	1.8	0.8	4.5	7.3	12.0	15.1	17.2	16.3	13.5	8.9	4.8	1.1	(33)	
		MCDB	4.5	3.2	9.3	14.0	19.3	22.5	25.8	25.9	22.9	15.0	9.2	3.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	10.7	11.0	12.3	12.8	12.4	14.4	14.2	13.3	11.9	10.4	9.8	(35)	
		MCDBR	10.0	12.0	14.1	16.8	16.7	16.0	17.7	17.9	18.2	15.5	12.9	10.3	(36)	
	5% WB	MCWBR	7.2	8.7	8.0	8.4	7.4	6.6	6.5	6.3	7.2	7.7	7.3	6.9	(37)	
		MCWBR	9.9	11.6	13.8	15.7	14.6	14.4	14.8	15.3	16.9	14.2	12.7	10.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.275	0.263	0.297	0.353	0.376	0.365	0.368	0.385	0.338	0.299	0.267	0.251	(40)	
		taud	2.352	2.471	2.441	2.287	2.268	2.361	2.366	2.290	2.400	2.490	2.481	2.417	(41)	
	Ebn at Noon		775	903	918	889	880	892	883	847	857	840	794	764	(42)	
		Edh at Noon	65	74	91	120	128	117	115	118	95	72	56	52	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.25	2.24	3.58	4.92	5.85	6.35	6.90	5.57	3.99	2.48	1.39	0.97	(44)	
	RadStd		0.12	0.19	0.25	0.34	0.41	0.39	0.36	0.37	0.33	0.25	0.10	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page